Given: Due:	Further Maths (Core Pure) HW 19	Name: MARK SCHEME

Chapter 4: Roots or Polynomials

1. The quadratic equation $2z^2 - 4z + 5 = 0$ has roots α and β.
   (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$. [2]
   (ii) Find the quadratic equation with roots $3\alpha - 1$, $3\beta - 1$. [3]
   (iii) Find the cubic equation which has roots α, β and $\alpha + \beta$. [4]

1. (i) $2z^2 - 4z + 5 = 0$

$$\alpha + \beta = -\frac{b}{a} = -\frac{-4}{2} = 2$$

$$\alpha\beta = \frac{c}{a} = \frac{5}{2}$$

[2]

(ii) Let $w = 3z - 1$, so $z = \frac{w+1}{3}$

Substituting into quadratic equation:

$$2\left(\frac{w+1}{3}\right)^2 - 4\left(\frac{w+1}{3}\right) + 5 = 0$$

$$2(w+1)^2 - 12(w+1) + 45 = 0$$

$$2w^2 + 4w + 2 - 12w - 12 + 45 = 0$$

$$2w^2 - 8w + 35 = 0$$

[3]

(iii) The original quadratic equation has roots α and β.

The value of $\alpha + \beta = 2$

The cubic equation is therefore $(2z^2 - 4z + 5)(z - 2) = 0$

$$2z^3 - 4z^2 + 5z - 4z^2 + 8z - 10 = 0$$

$$2z^3 - 8z^2 + 13z - 10 = 0$$

[4]

2. The equation $z^3 + kz^2 - 4z - 12 = 0$ has roots α, β and γ.
   (i) Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$, and express k in terms of α, β and γ. [3]
   (ii) For the case where $\gamma = -\alpha$, solve the equation and find the value of k. [4]
   (iii) For the case $k = 5$, find a cubic equation with roots $2 - \alpha$, $2 - \beta$, $2 - \gamma$. [4]

2. (i) $z^3 + kz^2 - 4z - 12 = 0$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = -4$

$\alpha\beta\gamma = -\frac{d}{a} = 12$

$\alpha + \beta + \gamma = -\frac{b}{a} = -k$

$k = -\alpha - \beta - \gamma$

[3]

(ii) If $\gamma = -\alpha$: $\alpha\beta - \beta\alpha - \alpha^2 = -4$

$\alpha^2 = 4$

$\alpha = \pm 2$

$-\alpha^2\beta = 12$

$-4\beta = 12$

$\beta = -3$

The roots are 2, -2 and -3

$k = -2 + 2 + 3 = 3$

[4]

(iii) $z^3 + 5z^2 - 4z - 12 = 0$

$w = 2 - z$ so $z = 2 - w$

Substituting into cubic equation:

$(2-w)^3 + 5(2-w)^2 - 4(2-w) - 12 = 0$

$8 - 12w + 6w^2 - w^3 + 20 - 20w + 5w^2 - 8 + 4w - 12 = 0$

$w^3 - 11w^2 + 28w - 8 = 0$

[4]

3. The complex number 3 + 2i is a root of the equation $2x^3 + px^2 + 20x + q = 0$, where p and q are real numbers.

(i) Find the other two roots of the cubic equation. [4]

(ii) Find the values of p and q. [4]

3. (i) $3 + 2i$ is a root, so $3 - 2i$ is also a root.

$$\sum \alpha\beta = \frac{c}{a}$$

$$(3+2i)\alpha + (3-2i)\alpha + (3+2i)(3-2i) = \frac{20}{2}$$

$$6\alpha + 13 = 10$$

$$6\alpha = -3$$

$$\alpha = -\tfrac{1}{2}$$

The other two roots are $3 - 2i$ and $-\tfrac{1}{2}$.

[4]

(ii) $$\sum \alpha = -\frac{b}{a}$$

$$3 + 2i + 3 - 2i - \tfrac{1}{2} = -\frac{p}{2}$$

$$5.5 = -\frac{p}{2}$$

$$p = -11$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$-\tfrac{1}{2}(3+2i)(3-2i) = -\frac{q}{2}$$

$$q = 13$$

[4]

4.

$$f(z) = z^3 - 8z^2 + pz - 24$$

where p is a real constant.

Given that the equation $f(z) = 0$ has distinct roots

$$\alpha, \beta \text{ and } \left(\alpha + \frac{12}{\alpha} - \beta\right)$$

(a) solve completely the equation $f(z) = 0$

(6)

(b) Hence find the value of p.

(2)

This question was very well answered compared to the other questions late in the paper, probably due to the fact the question was more familiar to candidates due to a similar question having appeared on the Specimen Assessment Materials. Student did seem more well prepared for this question, and full marks was achieved by over 40% of candidates.

(a) The required procedure of using the sum of roots to eliminate β was achieved by most candidates, albeit amongst similar equations for the pair sum and product. Writing out all the information was a common first step, rather than identifying only that which was needed. Only a very small number failed to successfully eliminate β.

Most could then efficiently multiply throughout by α to get a correct equation though some then made algebraic errors rearranging to a quadratic. Solving the quadratic equation was usually performed correctly. Going on to find the third root seemed more troublesome for many candidates, some of whom did not realise that they had found two different roots already in the conjugate pair. The quickest way of finding the third root was to use the sum of roots being equal to 8, but the fact that the product of all three is 24 was more often used (perhaps since this was the method exemplified in the SAMS). Other candidates used longer methods for finding the second and third roots, such as use of the pair sum where a quadratic for β was required to be solved, but such methods were more difficult to complete successfully.

(b) Most candidates had a correct method for find p, usually using the pair sum of roots, but slips at various points prevented some candidates from achieving full marks for the question. Also popular as a method was multiplying out the expression $(z - \alpha)(z - \beta)(z - \gamma)$, and given that α and β are conjugates this could be performed quite quickly for the astute student. Use of the factor theorem was the most direct method, but was used infrequently.

Question	Scheme	Marks	AOs
(a)	$\alpha+\beta+\left(\alpha+\frac{12}{\alpha}-\beta\right)=8$ so $2\alpha+\frac{12}{\alpha}=8$	M1	1.1b
		A1	1.1b
	$\Rightarrow 2\alpha^2-8\alpha+12=0$ or $\alpha^2-4\alpha+6=0$ $\Rightarrow \alpha=\frac{4\pm\sqrt{(-4)^2-4(1)(6)}}{2(1)}$ or $(\alpha-2)^2-4+6=0 \Rightarrow \alpha=...$	M1	1.1b
	$\Rightarrow \alpha=2\pm i\sqrt{2}$ are the two complex roots	A1	1.1b
	A correct full method to find the third root. Common methods are: Sum of roots $=8 \Rightarrow$ third root $=8-\left(2+i\sqrt{2}\right)-\left(2-i\sqrt{2}\right)=...$ third root $=2+i\sqrt{2}+\frac{12}{2+i\sqrt{2}}-\left(2-i\sqrt{2}\right)=...$ Product of roots $=24 \Rightarrow$ third root $=\frac{24}{\left(2+i\sqrt{2}\right)\left(2-i\sqrt{2}\right)}=...$ $(z-\alpha)(z-\beta)=z^2-4z+6 \Rightarrow f(z)=\left(z^2-4z+6\right)(z-\gamma)\Rightarrow \gamma=...$ (or long division to find third factor).	M1	3.1a
	Hence the roots of $f(z)=0$ are $2\pm i\sqrt{2}$ and 4	A1	1.1b
		(6)	
(b)	E.g. $f(4)=0 \Rightarrow 4^3-8\times4^2+4p-24=0 \Rightarrow p=...$ Or $p=\left(2+i\sqrt{2}\right)\left(2-i\sqrt{2}\right)+4\left(2+i\sqrt{2}\right)+4\left(2-i\sqrt{2}\right)\Rightarrow p=...$ Or $f(z)=(z-4)(z^2-4z+6)\Rightarrow p=...$	M1	3.1a
	$\Rightarrow p=22$ cso	A1	1.1b
		(2)	
		(8 marks)	

Notes		
(a)	M1	Equates sum of roots to 8 and obtains an equation in just α.
	A1	Obtains a correct equation in α.
	M1	Forms a three term quadratic equation in α and attempts to solve this equation by either completing the square or using the quadratic formula to give $\alpha=....$
	A1	$\alpha=2\pm i\sqrt{2}$
	M1	Any correct method for finding the remaining root. There are various routes possible. See scheme for common ones. Allow this mark if -24 is used as the product. See note below for a less common approach.
	A1	Third root found with all three roots correct. Note α and β need not be identified.
(b)	M1	Any correct method of finding p. For example, applies the factor theorem, process of finding the pair sum of roots, or uses the roots to form f(z).
	A1	$p=22$ by correct solution only. Note: this can be found using only their complex roots from (a) (e.g. by factor theorem)

Note for (a) final M – it is possible to find the second and third roots using only one initial root (e.g. if second root forgotten or error leads to only one initial root being found).

Product of roots $=\alpha\beta\left(\alpha+\frac{12}{\alpha}-\beta\right)=24 \Rightarrow \alpha\beta^2-\left(\alpha^2+12\right)\beta+24=0$, substitutes in α and attempts to solve the quadratic in β to achieve remaining roots. The final M can be gained once three roots in total have been obtained. (This is unlikely to be seen as part of a correct answer.) Allow if -24 has been used for the product.

5. The cubic equation $2z^3+pz^2+qz+r=0$ has roots $\frac{\alpha}{k},\alpha,k\alpha$.

(i) Express p, q and r in terms of k and α. [3]

(ii) Show that $2q^3=p^3r$. [3]

(iii) Solve the equation for the case where $p=q=-3$. [4]

5. (i) $2z^3 + pz^2 + qz + r = 0$

$$\sum \alpha = -\frac{b}{a}$$

$$\frac{\alpha}{k} + \alpha + k\alpha = -\frac{p}{2}$$

$$p = -2\alpha\left(\frac{1}{k} + 1 + k\right)$$

$$\sum \alpha\beta = \frac{c}{a}$$

$$\left(\frac{\alpha}{k} \times \alpha\right) + (\alpha \times k\alpha) + \left(k\alpha \times \frac{\alpha}{k}\right) = \frac{q}{2}$$

$$q = 2\alpha^2\left(\frac{1}{k} + k + 1\right)$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$\frac{\alpha}{k} \times \alpha \times k\alpha = -\frac{r}{2}$$

$$r = -2\alpha^3$$

[3]

(ii) Dividing the first two equations: $\frac{p}{q} = \frac{-2\alpha}{2\alpha^2}$

$$\alpha = -\frac{q}{p}$$

Substituting into third equation: $r = -2\left(-\frac{q}{p}\right)^3$

$$p^3 r = 2q^3$$

[3]

(iii) $p = q = -3 \quad \Rightarrow r = 2$

$r = -2\alpha^3 \quad \Rightarrow 2 = -2\alpha^3 \quad \Rightarrow \alpha = -1$

$$p = -2\alpha\left(\frac{1}{k} + 1 + k\right)$$

$$-3 = 2\left(\frac{1}{k} + 1 + k\right)$$

$$-3k = 2 + 2k + 2k^2$$

$$2k^2 + 5k + 2 = 0$$

$$(2k + 1)(k + 2) = 0$$

$k = -\frac{1}{2}$ or -2

The roots are $\frac{1}{2}$, -1 and 2.

[4]